

Let's work through a **complete numerical example** showing exactly how XGBoost chooses a split node using the **gain formula**.

To keep the mathematics manageable, we'll use:

- Regression problem
 - Squared error loss
 - One tree
 - Max depth = 2
 - $\lambda = 1$
 - $\gamma = 0$
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Training Data

Sample	x	y
1	1	10
2	2	12
3	3	14
4	4	20
5	5	22
6	6	24

Step 1: Initial Prediction

XGBoost starts with the mean value:

$$\hat{y} = \frac{10 + 12 + 14 + 20 + 22 + 24}{6}$$
$$= 17$$

Thus every sample is initially predicted as 17.

Step 2: Calculate Gradients and Hessians

For squared error

$$L = \frac{1}{2} (y - \hat{y})^2$$

Gradient:

$$g_i = \hat{y}_i - y_i$$

Hessian:

$$h_i = 1$$

Sample	y	Prediction	Gradient g
1	10	17	7
2	12	17	5
3	14	17	3
4	20	17	-3
5	22	17	-5
6	24	17	-7

All Hessians are 1.

Root Node

All samples belong to the root.

$$G = 7 + 5 + 3 - 3 - 5 - 7 = 0$$
$$H = 6$$

Root score:

$$Score_{root} = \frac{G^2}{H + \lambda} = \frac{0^2}{7} = 0$$

Step 3: Evaluate Candidate Splits

Possible splits:

$$x < 1.5$$

$$x < 2.5$$

$$x < 3.5$$

$$x < 4.5$$

$$x < 5.5$$

Candidate 1: $x < 1.5$

Left:

$$\{1\}$$

Right:

$$\{2, 3, 4, 5, 6\}$$

Left

$$G_L = 7$$

$$H_L = 1$$

Right

$$G_R = -7$$

$$H_R = 5$$

Gain:

$$\frac{1}{2} \left(\frac{7^2}{2} + \frac{(-7)^2}{6} - 0 \right)$$

$$\begin{aligned}
 &= \frac{1}{2}(24.5 + 8.17) \\
 &= 16.33
 \end{aligned}$$

Candidate 2: $x < 2.5$

Left:

$$\{1, 2\}$$

Right:

$$\{3, 4, 5, 6\}$$

Left

$$\begin{aligned}
 G_L &= 12 \\
 H_L &= 2
 \end{aligned}$$

Right

$$\begin{aligned}
 G_R &= -12 \\
 H_R &= 4
 \end{aligned}$$

Gain:

$$\begin{aligned}
 &\frac{1}{2}\left(\frac{144}{3} + \frac{144}{5}\right) \\
 &= \frac{1}{2}(48 + 28.8) \\
 &= 38.4
 \end{aligned}$$

Candidate 3: $x < 3.5$

Left:

$$\{1, 2, 3\}$$

Right:

$$\{4, 5, 6\}$$

Left

$$G_L = 15$$

$$H_L = 3$$

Right

$$G_R = -15$$

$$H_R = 3$$

Gain:

$$\frac{1}{2} \left(\frac{225}{4} + \frac{225}{4} \right) \\ = 56.25$$

Candidate 4: $x < 4.5$

Left:

$$G_L = 12$$

$$H_L = 4$$

Right:

$$G_R = -12$$

$$H_R = 2$$

Gain:

$$38.4$$

Candidate 5: $x < 5.5$

Gain:

16.33

Root Split Selection

Split	Gain
$x < 1.5$	16.33
$x < 2.5$	38.40
$x < 3.5$	56.25
$x < 4.5$	38.40
$x < 5.5$	16.33

Maximum gain:

56.25

Therefore the root node becomes

$x < 3.5$

This is how XGBoost chooses the first split.

Level 2: Split Left Child

Current left node:

x g

1 7

2 5

x g

3 3

Node statistics:

$$G = 15$$

$$H = 3$$

Possible splits:

$$x < 1.5$$

$$x < 2.5$$

Split at $x < 1.5$

Left:

$$G_L = 7, H_L = 1$$

Right:

$$G_R = 8, H_R = 2$$

Gain:

$$\begin{aligned} & \frac{1}{2} \left(\frac{49}{2} + \frac{64}{3} - \frac{225}{4} \right) \\ &= \frac{1}{2} (24.5 + 21.33 - 56.25) \\ &= -5.21 \end{aligned}$$

Split at $x < 2.5$

Left:

$$G_L = 12, H_L = 2$$

Right:

$$G_R = 3, H_R = 1$$

Gain:

$$\begin{aligned} & \frac{1}{2} \left(\frac{144}{3} + \frac{9}{2} - 56.25 \right) \\ &= \frac{1}{2} (-3.75) \\ &= -1.875 \end{aligned}$$

Both gains are negative.

Therefore:

No split is performed.

The left node becomes a leaf.

Level 2: Split Right Child

Right child:

x g

4 -3

5 -5

6 -7

Statistics:

$$\begin{aligned} G &= -15 \\ H &= 3 \end{aligned}$$

Possible splits:

$$x < 4.5$$

$$x < 5.5$$

x<4.5

Left:

$$G_L = -3$$

Right:

$$G_R = -12$$

Gain:

$$-1.875$$

x<5.5

Left:

$$G_L = -8$$

Right:

$$G_R = -7$$

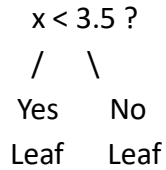
Gain:

$$-5.21$$

Again both gains are negative.

No split occurs.

Final Tree



Compute Leaf Values

For left leaf:

$$\begin{aligned} G &= 15 \\ H &= 3 \end{aligned}$$

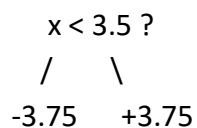
Optimal weight:

$$\begin{aligned} w &= -\frac{15}{3 + 1} \\ &= -3.75 \end{aligned}$$

For right leaf:

$$\begin{aligned} G &= -15 \\ H &= 3 \\ w &= -\frac{-15}{4} \\ &= 3.75 \end{aligned}$$

Final Tree



The key idea is that **at every node, XGBoost evaluates every possible split, computes the gain using gradient and Hessian statistics, and selects the split with the highest positive gain.** If no candidate split yields a positive gain, the node becomes a leaf. This greedy gain-maximization process is repeated level by level until the maximum depth or stopping criteria are reached.